Euclid's Elements — Book I
Proposition 37

Formal Reconstruction — Technical Documentation

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Triangles which are on the same base and in the same parallels are equal to one another.

1 Introduction

Proposition I.37 is the first triangle theorem in Euclid's area block. Two triangles have the same base BC, while their third vertices A and D lie on one line parallel to BC. Euclid concludes that the two triangles are "equal." At this point of Book I, as already in I.35 and I.36, this is an equality of content rather than congruence of triangles.

The formal reconstruction keeps that conclusion synthetic. It introduces no numerical area and does not strengthen the result to unconditional finite equidecomposability. The public Lean theorem concludes `HilbertScissorsEquicomplementable`.

The principal change from Euclid's proof is the treatment of the final "halving" step. Euclid surrounds each triangle by a parallelogram, proves the two parallelograms equal by I.35, and then appeals to the fact that halves of equals are equal. The present reconstruction does not add a general cancellation or division-by-two principle. Instead it follows Hilbert's Chapter IV organization: each triangle is first converted, by the midpoint construction of Hilbert's Theorem 45, into an equidecomposable parallelogram; those two parallelograms are then compared by the already formalized I.35 scissors machinery.

This makes I.37 a particularly clear example of the three layers that now have to be kept distinct:

$$\text{geometry} \quad \longrightarrow \quad \text{representation} \quad \longrightarrow \quad \text{content.}$$

2 The Classical Configuration

Let ABC and DBC be triangles on the same base BC. Their top vertices A and D lie on a line parallel to BC:

$$AD \parallel BC.$$

Euclid says that the two triangles are in the same parallels AD and BC.

The current Lean theorem encodes the configuration with the single strict parallelism hypothesis

$$\mathrm{Parallel}(A, D; B, C).$$

Because the project's parallel relation is nondegenerate, this already gives $A \neq D$ and $B \neq C$, and the disjointness of the two carrier lines forces both A and D off the base line. Thus the two triangles are internally proved noncollinear; nondegeneracy is not an extra theorem hypothesis.

The formal proof then constructs four midpoint witnesses:

$$\begin{array}{ll} M \text{ midpoint of } BA, & N \text{ midpoint of } CA, \\ P \text{ midpoint of } BD, & Q \text{ midpoint of } CD. \end{array}$$

Hilbert's Theorem 45 is applied to the two triangles and produces points F and G such that

$$MBCF \quad \text{and} \quad PBCG$$

are parallelograms, while the original triangles are scissors-equivalent to these parallelogram terms.

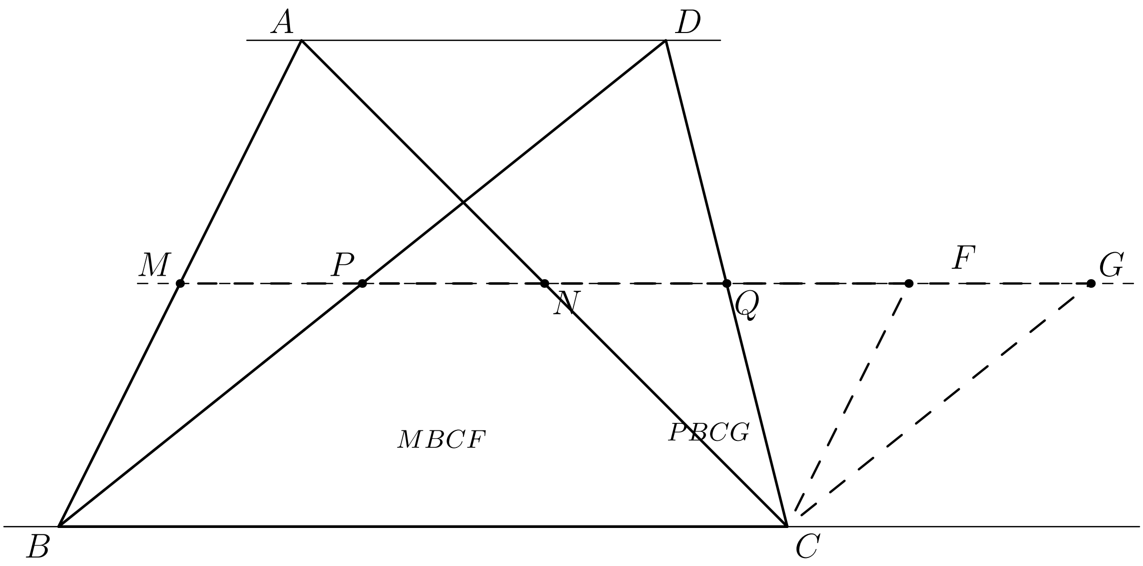

Figure 1: The geometric reduction used by the Lean proof. The solid triangles ABC and DBC are the figures in the public theorem. The points M, N, P, Q are the four midpoint witnesses used by two applications of Hilbert's Theorem 45; the dashed figures $MBCF$ and $PBCG$ are the resulting parallelograms. The displayed order $M - P - F - G$ is one representative strict case only. The proof derives the common upper carrier and handles all possible orders rather than assuming the order shown in the picture.

3 Classical Proof

Euclid extends the top line AD in both directions to points E and F. Through B he draws a line parallel to CA, and through C a line parallel to BD, using I.31. This produces two parallelograms

$$EBCA \qquad \text{and} \qquad DBCF.$$

They have the same base BC and lie in the same parallels, so I.35 gives

$$EBCA = DBCF.$$

By I.34 the diagonal AB bisects the first parallelogram into two congruent triangles, and the diagonal DC bisects the second. Hence

$$\triangle ABC \text{ is half of } EBCA, \qquad \triangle DBC \text{ is half of } DBCF.$$

Euclid then concludes that the two halves are equal.

The classical dependency graph is therefore

$$\begin{gathered} AD \parallel BC \\ \Downarrow \quad \text{I.31 twice} \\ EBCA,\ DBCF \text{ parallelograms} \\ \Downarrow \quad \text{I.35} \\ EBCA = DBCF \\ \Downarrow \quad \text{I.34 twice} \\ \triangle ABC,\ \triangle DBC \text{ are corresponding halves} \\ \Downarrow \quad \text{“halves of equals are equal”} \\ \triangle ABC = \triangle DBC. \end{gathered}$$

The final step is the important foundational point. In Heath's presentation the principle "halves of equal things are equal" is bracketed: it is not one of Euclid's explicitly stated Common Notions. A formal development must therefore either justify a suitable cancellation principle or avoid the step. The present reconstruction takes the second route.

4 Hilbert's Theorems 44–46 and the Halving Issue

Hilbert's Chapter IV gives a direct synthetic organization of exactly the material needed here.

Theorem 44. Parallelograms with equal bases and equal altitudes are *equicomplementable*.

Theorem 45. Every triangle is *equidecomposable* with a parallelogram having an equal base and half the altitude. Hilbert's construction bisects two sides of the triangle and extends the segment joining the midpoints by an equal amount. The two small triangles created by the construction are congruent, so the triangle can be rearranged into the parallelogram.

Theorem 46. From Theorems 44 and 45, together with transitivity of the content relation, Hilbert obtains that triangles with equal bases and equal altitudes are equicomplementable.

This is the correct control theorem for I.37. Hilbert also explicitly warns that the stronger assertion of finite equidecomposability is not available in full generality without an Archimedean hypothesis: in non-Archimedean geometries triangles of equal base and altitude can be equicomplementable without being equidecomposable.

Hartshorne makes the same logical issue visible in the language of equal content. In Section 22 he observes that Euclid's original proof of I.37 uses "halves of equals are equal," a property depending on his additional property (Z). He therefore gives a midpoint/parallelogram proof of I.37 which avoids (Z).

The Lean reconstruction follows the same conceptual escape from cancellation, but its exact proof graph is Hilbertian: Theorem 45 is formalized directly in the scissors calculus, and I.35

supplies the parallelogram content comparison. No general theorem of the form

$$2P \sim 2Q \quad \Longrightarrow \quad P \sim Q$$

is introduced.

5 Proof Architecture of the Reconstruction

5.1 Step 1: nondegeneracy from the parallel hypothesis

The helper

```
i37_nondegenerate_triangles
```

derives

$$A, B, C \text{ noncollinear}, \qquad D, B, C \text{ noncollinear}, \qquad B, A, D \text{ noncollinear}.$$

The proof uses the disjoint carriers contained in the strict parallelism hypothesis. Thus the ordinary diagrammatic statement that the top vertices are off the base is converted into explicit Hilbert incidence information.

5.2 Step 2: four midpoint witnesses and Hilbert 45 twice

The helper

```
i37_half_parallelograms
```

constructs M, N, P, Q as the four midpoints shown in Figure 1. It then applies

```
hilbert_triangle_to_parallelogram
```

twice.

For triangle ABC the internal call is made in the cyclic presentation BCA and returns a point F with

$$BCFM \text{ a parallelogram}$$

and a formal scissors equality between the triangle and that parallelogram. For triangle DBC the analogous construction returns G and the parallelogram $BCGP$.

This theorem is the formal version of Hilbert's Theorem 45. No area value is computed. The result is a finite triangle-rearrangement certificate in `HilbertScissorsEq`.

5.3 Step 3: the common upper carrier

The midpoint theorem is applied in triangle BAD. Since M is the midpoint of BA and P is the midpoint of BD,

$$MP \parallel AD.$$

The public hypothesis gives $AD \parallel BC$, and the developed Euclidean parallel theory yields

$$MP \parallel BC.$$

The two Hilbert-45 parallelograms also give

$$FM \parallel BC, \qquad GP \parallel BC.$$

Uniqueness/transitivity of parallels then identifies their carriers with the carrier MP. Hence

$$M, F, P \text{ are collinear,} \qquad M, P, G \text{ are collinear.}$$

This is packaged by

```
i37_upper_line
```

and is one of the most important hidden configuration facts of the proof.

5.4 Step 4: equal upper sides

Opposite-side congruence in the two parallelograms gives

$$MF \cong BC, \qquad PG \cong BC.$$

Therefore

$$MF \cong PG.$$

The helper

```
i37_upper_sides_congruent
```

packages this short metric step.

5.5 Step 5: representation normalization

Hilbert 45 naturally returns parallelograms in the forms $BCFM$ and $BCGP$. Proposition I.35 is most conveniently applied to the cyclic presentations $MBCF$ and $PBCG$, in which BC is the common base and the upper sides are MF and PG.

The helper

```
i37_parallelogram_reduction
```

performs precisely this normalization. The geometric parallelograms are rotated by

```
ParallelogramRotateOne
```

and the corresponding formal scissors terms are transported by

```
parallelogram_term_rotateOne
```

This step changes no geometry; it only changes the representation expected by the downstream API.

At the end of the reduction we have points M, P, F, G satisfying

$$\begin{gathered} MBCF,\ \ PBCG \text{ parallelograms}, \\ MP \parallel BC, \\ M, F, P \text{ collinear}, \qquad M, P, G \text{ collinear}, \\ MF \cong PG, \end{gathered}$$

together with scissors equivalences

$$\triangle ABC \sim MBCF, \qquad \triangle DBC \sim PBCG.$$

Here $\sim$ in this display denotes `HilbertScissorsEq`. The stronger content relation used later is

```
HilbertScissorsEquicomplementable
```

5.6 Step 6: order normalization on the common upper line

Proposition I.35 is sensitive to the actual order of the four upper points. The formal proof therefore does not read that order from Figure 1. It first handles two boundary configurations:

$$M = P, \qquad F = P.$$

If $M = P$, uniqueness of the fourth vertex of a parallelogram gives $F = G$, so the two reduced parallelograms coincide. This is handled by

```
i37_finish_same_left_vertex
```

using the reusable theorem

```
ParallelogramFourthVertexUnique
```

If $F = P$, the two parallelograms have a common upper endpoint, and the existing I.35 common-endpoint helper closes the content comparison. This is

```
i37_finish_common_upper_endpoint
```

In the remaining case, Hilbert's betweenness trichotomy gives exactly three strict orders of M, F, P:

$$M - F - P, \qquad F - M - P, \qquad M - P - F.$$

The helper

```
i37_upper_trichotomy
```

packages this split.

5.7 Step 7: the orientation lemma for the strict cases

The case $M - F - P$ contains the deepest proposition-local order argument. The helper

```
i37_order_MFP_implies_FPG
```

proves

$$M - F - P \quad \Longrightarrow \quad F - P - G.$$

The proof is genuinely synthetic. In triangle PMB, the line CF enters through the side PM at F. Since $CF \parallel MB$, it cannot leave through MB, so Pasch forces an intersection with PB. The existing helper `hilbert_collinear_between_of_parallel` locates this crossing on FC. The second parallelogram supplies SameSide information for C and G relative to the line PB. Hence F and G are on opposite sides of PB. Their segment must cross PB, and incidence uniqueness identifies that crossing point with P. Thus $F - P - G$.

This is exactly the sort of positional statement that is obvious in a well-drawn affine picture but cannot be taken from the picture in a Hilbert formalization.

The case $F - M - P$ is closed by reversing the two parallelogram presentations and applying I.35 in the reversed configuration. The helper is

```
i37_finish_order_FMP
```

For the case $M - P - F$, the helper

```
i37_order_MPF_implies_PFG
```

uses the congruence $MF \cong PG$, strict segment comparison, and a reuse of the previous orientation lemma to eliminate the two impossible orders of P, F, G. The surviving order is

$$P - F - G.$$

The final content step for this branch is packaged in

```
i37_finish_order_MPF
```

5.8 Step 8: Proposition I.35 and transport back to the triangles

Once the correct upper order is known, Proposition I.35 compares the two reduced parallelograms:

$$MBCF \quad \text{and} \quad PBCG.$$

The resulting relation is `HilbertScissorsEquicomplementable`.

Finally

```
equicomplementable_transport
```

uses the two Hilbert-45 scissors equivalences to move the conclusion from the parallelogram representatives back to the original triangles. This is the content stage of the proof; the preceding midpoint, parallel, Pasch, and order arguments belong to the geometry stage.

6 Lean Formalization

The current file imports

```
import CGJteamLab.Proposition35
import CGJteamLab.MidsegmentParallel
```

Thus I.35 is the direct Book I content dependency, while the midpoint theorem is imported as developed Hilbert infrastructure. Proposition I.36 is not a formal dependency of I.37.

The public theorem is

```
theorem euclid_proposition_37
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C D : Geo.Point)
    (hAD_BC : Geo.Parallel A D B C) :
    HilbertScissorsEquicomplementable Geo
      (hilbertScissorsTriangle Geo A B C)
      (hilbertScissorsTriangle Geo D B C)
```

The proposition-local proof path is represented by the following declarations:

```
i37_nondegenerate_triangles
i37_half_parallelograms
i37_upper_line
i37_upper_sides_congruent
i37_parallelogram_reduction
i37_finish_from_right_cases
i37_finish_same_left_vertex
i37_finish_common_upper_endpoint
i37_upper_trichotomy
i37_finish_order_FMP
i37_finish_order_MFP
i37_order_MFP_implies_FPG
i37_order_MPF_implies_PFG
i37_finish_order_MPF
euclid_proposition_37
```

At the top level the final theorem is intentionally short. It first calls

```
i37_parallelogram_reduction
```

then separates the two boundary cases, uses

```
i37_upper_trichotomy
```

for the three strict cases, and dispatches each case to its corresponding finishing helper.

The crucial point is that the top-level theorem contains no hidden arithmetic area operation and no cancellation axiom. The only equality-of-figures relations in the statement and final transport are the formal scissors relations already introduced at I.35.

7 Relationship with Earlier Propositions

7.1 Proposition I.31

I.31 belongs to Euclid's literal construction. Euclid draws two parallels in order to surround the two triangles by parallelograms. The final Lean proof does not call Proposition I.31 by name. Its construction is replaced by Hilbert's midpoint theorem and Hilbert 45.

7.2 Proposition I.34

Classically I.34 supplies the fact that a diagonal bisects a parallelogram and therefore that each original triangle is one half of its surrounding parallelogram. This is precisely the source of the final halving issue.

The Lean reconstruction does not use I.34 as a Book I theorem to divide equal parallelograms by two. Its relevant parallelogram congruence content is already available in the developed interface, while Hilbert 45 provides a direct scissors equivalence between each triangle and a smaller parallelogram.

7.3 Proposition I.35

I.35 is the decisive earlier proposition in the actual Lean dependency graph. After both triangles have been converted to parallelogram terms and their upper-order configuration has been normalized, I.35 proves those parallelograms equicomplementable. The result is then transported back to the triangles.

Thus the common core of Euclid and Lean is still I.35, but the bridge from triangles to parallelograms is different.

7.4 Proposition I.36

I.36 is historically adjacent but is not used by the final formal proof. It compares parallelograms on equal bases, whereas I.37 has a common base from the start. The two propositions nevertheless confirm the same architectural lesson: the content machinery belongs in a reusable scissors layer, while the new work in each proposition is geometric normalization.

8 Hilbert and Book Zero Components Used

The proof uses a broad but well-separated collection of reusable results.

8.1 Midpoint and midsegment infrastructure

The midpoint layer supplies

- midpoint existence;
- midpoint symmetry;
- conversion of midpoint data into strict betweenness and congruence;
- `hilbert_midpoints_noncollinear`;
- `MidsegmentTheorem`.

These results produce the four midpoint witnesses and the parallel line MP.

8.2 Hilbert Theorem 45 in the scissors layer

The reusable theorem

```
hilbert_triangle_to_parallelogram
```

is the central new content bridge. Its geometric construction is provided by

```
hilbert_triangle_parallelogram_data
```

and its scissors conclusion states that the triangle term is `HilbertScissorsEq` to the corresponding parallelogram term.

This theorem was introduced while reconstructing I.37 but belongs in the reusable scissors/interface layer rather than in the proposition-local file.

8.3 Parallel and parallelogram infrastructure

The proof uses

- transport of parallelism along collinear carriers;
- Euclidean transitivity/uniqueness of parallels;
- `ParallelogramOppositeSidesCongruent`;
- `ParallelogramFourthVertexUnique`;
- `ParallelogramRotateOne`;
- representation-level parallelogram rotations in the scissors layer.

8.4 Order, Pasch, and segment comparison

The strict-case normalization uses

- Hilbert betweenness trichotomy;
- inner and outer betweenness transitivity;
- Pasch's axiom through the derived intersection machinery;
- SameSide and OppositeSide transport;
- incidence uniqueness of the intersection of two distinct lines;
- strict segment comparison and its transport under congruence.

No new Book Zero theorem is introduced by the final Proposition file. The proof instead exercises the existing Book Zero/order language and adds several reusable helpers to the Interface and Scissors layers.

9 Formalization Notes

9.1 Geometry, representation, and content

I.37 makes the three-layer architecture especially explicit.

Geometry. The geometric work constructs midpoint data, obtains two parallelograms, proves that their upper sides lie on one common carrier, and determines the possible point orders on that carrier. Pasch and side-of-line arguments occur only here.

Representation. The same geometric parallelogram can appear as $BCFM$, $MBCF$, $FCBM$, and other cyclic or reversed presentations. Lean cannot silently identify these argument lists. Helpers such as `ParallelogramRotateOne`, `parallelogram_term_rotateOne`, and the reversal lemmas normalize the presentation without adding geometric content.

Content. Once the representatives are aligned, I.35 and `equicomplementable_transport` are sufficient. No new area function, polygon semantics, or cancellation rule is introduced.

9.2 Hidden configuration data

The proof explicitly derives or manages facts that the classical diagram suppresses:

- noncollinearity of both original triangles;
- four strict midpoint betweenness relations;
- the midpoint parallel $MP \parallel AD$;
- the derived common upper carrier $MP \parallel BC$;

- collinearity of M, F, P, G on that carrier;
- congruence $MF \cong PG$;
- the boundary cases $M = P$ and $F = P$;
- the three strict orders $M - F - P$, $F - M - P$, $M - P - F$;
- Pasch data in triangle PMB;
- SameSide/OppositeSide information relative to PB;
- the exact crossing point used to prove $F - P - G$;
- strict segment comparisons used to eliminate impossible orders in the $M - P - F$ branch.

None of these order relations is inferred merely from the chosen drawing.

9.3 Axiomatic status

The public theorem assumes

```
[HilbertEuclideanPlane Geo]
```

and is therefore Euclidean in the current project API.

The Euclidean parallel axiom enters through already developed infrastructure, not through a new I.37-specific axiom. In particular, the proof uses the midsegment module and Euclidean transitivity/uniqueness of parallels, and it calls Proposition I.35, whose parallelogram/content infrastructure is also exposed under the Euclidean-plane instance.

This documentation does not claim that every internal sublemma has the minimal possible axiom signature. The relevant project-level statement is that the completed I.37 theorem is derived from the existing `HilbertEuclideanPlane` hierarchy and introduces no new geometric axiom. No continuity axiom is added at I.37.

9.4 Why the conclusion is equicomplementability

The conclusion is deliberately

```
HilbertScissorsEquicomplementable
```

rather than `HilbertScissorsEq`. This matches Hilbert's Theorem 46 and avoids an invalid strengthening in non-Archimedean settings. The individual Hilbert-45 reductions are scissors equivalences, but the comparison of the two resulting parallelograms comes from I.35 and therefore has the weaker, correct content relation.

9.5 Reusable helpers introduced around I.37

The formalization produced reusable material beyond the proposition-local case analysis.

In `HilbertInterface`:

```
hilbert_triangle_parallelogram_data
hilbert_midpoints_noncollinear
ParallelogramFourthVertexUnique
ParallelogramRotateOne
```

In `HilbertScissors`:

```
hilbert_triangle_to_parallelogram
equicomplementable_transport
parallelogram_term_rotateOne
```

The declarations beginning with `i37_` remain proposition-local because they package the particular reduction and order analysis required by I.37.

9.6 Source audit

The source audit separates four functions of the literature.

Euclid, Heath, and Wilkins control the historical statement and classical DAG: I.31 constructs the two parallelograms, I.35 compares them, I.34 identifies the triangles as halves, and the proof ends with the unstated halving principle.

Hilbert, Chapter IV, Sections 18–19, controls the semantics and the preferred formal architecture. Theorems 44–46 distinguish equidecomposability from equicomplementability and supply the midpoint route from triangles to parallelograms.

Hartshorne, Section 22, independently identifies the halving step as depending on property (Z) and gives a proof of I.37 that avoids (Z) by using midpoints and parallelograms. His treatment also confirms that equal content must not be silently replaced by numerical area or by unconditional finite equidecomposability.

Borsuk–Szmielew and Greenberg remain control sources for the underlying order, midpoint, parallelogram, and Euclidean parallel infrastructure. Neither is the primary source for the equal-content semantics at this stage: that role is filled more directly by Hilbert and Hartshorne.

The Beeson proof corpus remains useful as an audit of hidden configuration conditions and Euclidean gaps, but the final I.37 architecture is not organized around Tarski-style tactics or a literal reproduction of Beeson's proof.

10 Dependency Summary

The actual formal dependency chain is

$$\begin{gathered}\text{Hilbert incidence, order, congruence, and Euclidean parallel axioms}\\ \Downarrow \\ \text{HilbertInterface / Book Zero order, midpoint, side-of-line theory}\\ \Downarrow \\ \text{parallelogram infrastructure + MidsegmentTheorem}\\ \Downarrow \\ \text{HilbertScissors + Hilbert Theorem 45}\\ \Downarrow \\ \texttt{i37\_parallelogram\_reduction}\\ \Downarrow \\ \text{boundary cases + upper-line order normalization}\\ \Downarrow \\ \text{Proposition I.35 on the reduced parallelograms}\\ \Downarrow \\ \texttt{equicomplementable\_transport}\\ \Downarrow \\ \texttt{euclid\_proposition\_37}.\end{gathered}$$

The historical and formal DAGs can be compared succinctly as

$$\begin{aligned}\text{Euclid} &: \text{I.31 twice} \to \text{I.35} \to \text{I.34 twice} \to \text{halving},\\ \text{Lean} &: \text{midpoints} \to \text{Hilbert 45 twice} \to \text{order geometry} \to \text{I.35} \to \text{transport}.\end{aligned}$$

New geometric axiom:	no
New proposition-local helpers:	yes
New reusable Interface/Scissors helpers:	yes
New Book Zero theorem:	no
New numerical area or polygon semantics:	no
Public theorem compiles:	yes
Full `lake build` re-run in this documentation pass:	no
`sorry` in `Proposition37.lean`:	no

11 Conclusion

Proposition I.37 is not merely I.35 with triangles substituted for parallelograms. It is the point where the classical phrase "half of an equal figure" becomes a genuine foundational obligation.

The reconstruction resolves that obligation without adding a division or cancellation axiom. Hilbert's Theorem 45 converts each triangle into an appropriate parallelogram by a finite congruent-piece rearrangement. The midsegment theorem and Euclidean parallel theory place the two resulting parallelograms on one common upper carrier. A substantial but localized order analysis then supplies the precise configuration needed by I.35. Finally the existing scissors transport returns the equicomplementability result to the original triangles.

The proof therefore makes the architecture of the area block more precise: geometry determines where the pieces and carriers lie, representation helpers put the same figures into the argument order required by the API, and the scissors layer alone carries equality of content. That separation is likely to be the reusable pattern for the propositions immediately following I.37.